

Simple Design Method of Two-Degree-of-Freedom PID Position Controller Based on Pole-Zero-Assignment for Linear Servo Motors

Hideaki Hirahara*, Kenshiro Kuroki** and Shu Yamamoto*

* Faculty of Human Resources Development, Polytechnic University, Tokyo, Japan

** Graduate Course of Master of Science in Manufacturing Engineering, Polytechnic University, Tokyo, Japan

Abstract—This paper presents an improved design method of Two-Degree-Of-Freedom (TDOF) PID position controller based on pole-zero-assignment suitable for Linear Servo Motor (LSM) drives. This method can be applied to both positioning and tracking drives by simply specifying pole-angle in the s-plane. Furthermore, the disturbance suppression characteristic can be adjusted by changing the crossover frequency of sensitivity and complementary sensitivity functions.

Simulation and experimental results on a vector-controlled LSM validate the proposed method.

Index Terms— linear servo motor, PID controller, pole-zero-assignment, two-degree-of-freedom

I. INTRODUCTION

Position control systems using LSM are used for various applications that require high-speed and high-accuracy performance, such as transport device and machine tools in factories. In these drive systems, a PID controller is generally employed. Among the PID controllers, the TDOF PID controller is well known as a controller that can design the disturbance and reference responses almost independently [1]. This controller has been often utilized in process control systems [2]-[3] since design strategy is clear and intelligible; PID parameters (K_P , K_I and K_D) are first tuned so that the disturbance response is optimized, and the reference response is improved by means of TDOF parameters (α and β) that have no effect on the disturbance response. However, most servo systems needs to maximize the performance of the reference response rather than that of the disturbance response. So, design methods for applying this controller to servo control systems are not so much. In [4]-[5], the adoption of these controllers in mechanical servo systems has been studied. However, these design methods are not applicable to positioning and tracking servo systems because the reference response is not always optimized.

Against this background, some researches have been conducted into how to design TDOF PID controllers for servo systems using model-based adaptive control

algorithms and pole-zero-assignment [6]-[7].

The authors have improve the pole-zero-assignment in [7] and studied a design method for a TDOF PID controller suitable for both positioning and tracking drives [8]-[12]. These design methods entail easily determining all parameters (K_P , K_I , K_D , α and β) from six items: mover mass, maximum load mass, thrust constant, cutoff frequency of position control system, crossover frequency of sensitivity and complementary sensitivity functions, pole-angle in the s-plane. Consequently, one can design a suitable TDOF PID controller for both positioning and tracking drives by changing over the pole-angle in the s-plane.

In a recent study [12], the authors have proposed a new pole-zero-assignment; an upgraded version of the pole-zero-assignment in [11]. The characteristics of this pole-zero-assignment have higher position responses and disturbance suppression characteristics than the pole-zero-assignment in [11] in tracking servo drives. However, it has not been verified by an actual machine. In addition, a method for adjusting the disturbance suppression characteristic has not been discussed yet.

In this paper, the characteristics of pole-zero assignments in [11] and [12] are compared, and the validity of [12] is demonstrated by actual LSM drives. Furthermore, it is shown that the disturbance suppression characteristic can be adjusted by changing the crossover frequency of sensitivity and complementary sensitivity functions. The proposed method is verified with simulation and experimental results.

II. TDOF PID POSITION CONTROL SYSTEM

Fig. 1 shows a TDOF PID position controller for LSM drives. Here M is the total mass of the motor mover ($= m + m_L$, m and m_L are the mover mass [kg] and loaded-work

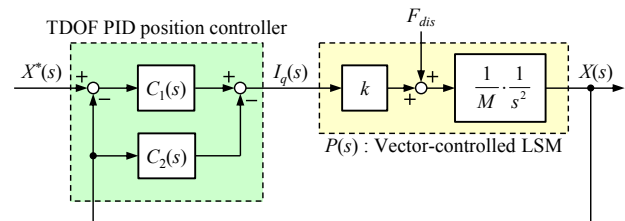


Fig. 1. Block diagram of a TDOF PID position control system.

This work was supported by Japan Society for the Promotion of Science (JSPS) KAKENHI Grant Number 20K04433.

mass [kg]), k is the thrust constant [N/A], x and x^* are the mover position [m] and command [m], $i_q(t)$ is the q-axis current [A], F_{dis} is the disturbance [N].

In this figure, transfer functions of controllers ($C_1(s)$ and $C_2(s)$) and a vector-controlled LSM ($P(s)$) are written as follows.

$$C_1(s) = (1 - \alpha)K_p + K_I \cdot \frac{1}{s} + (1 - \beta)K_D \cdot s, \quad (1)$$

$$C_2(s) = \alpha K_p + \beta K_D \cdot s, \quad (2)$$

$$P(s) = \frac{k}{M} \cdot \frac{1}{s^2}. \quad (3)$$

In addition, the closed-loop transfer function of the position control system $G_C(s)$ and sensitivity function $S(s)$ are obtained as follows.

$$G_C(s) = \frac{X(s)}{X^*(s)} = \frac{C_1(s)P(s)}{1 + (C_1(s) + C_2(s))P(s)}$$

$$= K(1 - \beta) \frac{s^2 + \frac{(1 - \alpha)q_1}{(1 - \beta)}s + \frac{q_1q_2}{(1 - \beta)}}{s^3 + Ks^2 + Kq_1s + Kq_1q_2}, \quad (4)$$

$$S(s) = \frac{1}{1 + (C_1(s) + C_2(s))P(s)}$$

$$= \frac{s^3}{s^3 + Ks^2 + Kq_1s + Kq_1q_2}, \quad (5)$$

where $q_1 = K_p / K_D$, $q_2 = K_I / K_p$, $K = kK_D / M$.

III. SIMPLE DESIGN METHOD OF TDOF PID CONTROLLER FOR LSM DRIVES

This chapter describes the design method proposed in [12]. This method is an upgraded version of pole-zero-assignment in [11].

Figs. 2 and 3 show the pole-zero-assignment in [11] and [12], respectively. In these figures, ω_b is the cutoff frequency of the closed-loop position control system, ε is an arbitrary constant ($\varepsilon > 1$), θ is the pole-angle in the s-plane.

The transfer function $G_C(s)$ with the poles $\{-\varepsilon\omega_b, -\omega_b(1 \pm j \tan \theta)\}$ and zeroes $\{-\varepsilon\omega_b, -\omega_b\}$ is written as following:

$$G_C(s) = \frac{A(s + \varepsilon\omega_b)(s + \omega_b)}{(s + \varepsilon\omega_b)\{s + \omega_b + j\omega_b \tan \theta\}\{s + \omega_b - j\omega_b \tan \theta\}}$$

$$= A \frac{s^2 + (\varepsilon + 1)\omega_b s + \varepsilon\omega_b^2}{s^3 + (\varepsilon + 2)\omega_b s^2 + (2\varepsilon + \delta)\omega_b^2 s + \varepsilon\delta\omega_b^3}, \quad (6)$$

where A is constant, $\delta = 1 + \tan^2 \theta$.

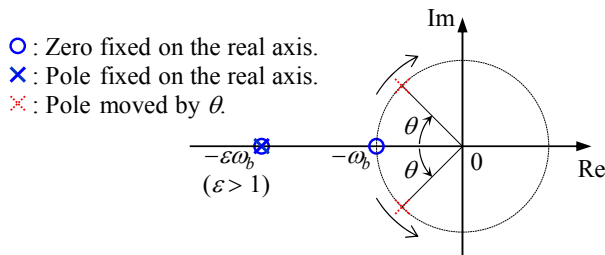


Fig. 2. Pole-zero-assignment in [11].

These design methods have the feature that the transient response can be adjusted by changing θ . In Fig. 2, the conjugate complex poles move along the arc according to the change of θ . In Fig. 3, however those poles move along the straight line on the real part ($-\omega_b$). As a result, even if θ changes, the real part of the poles does not vary and is kept in $-\omega_b$. It is considered that this results in higher position responses and disturbance suppression characteristics than the pole-zero-assignment shown in Fig. 2 in tracking servo drives.

The disturbance suppression characteristic can be adjusted by changing ε .

On the basis of the pole-zero-assignment shown in Fig. 3, the calculation formulae to obtain K_p , K_I , K_D , α and β can be derived so that the coefficients of Eq. (4) correspond to those of Eq. (6), which are summarized as follows.

$$K_p = \frac{M}{k} (2\varepsilon + \delta)\omega_b^2, \quad (7)$$

$$K_I = \frac{M}{k} \varepsilon \delta \omega_b^3, \quad (8)$$

$$K_D = \frac{M}{k} (\varepsilon + 2)\omega_b, \quad (9)$$

$$\alpha = \frac{\varepsilon(2 - \delta)}{2\varepsilon + \delta}, \quad (10)$$

$$\beta = \frac{\varepsilon - \delta + 2}{\varepsilon + 2}. \quad (11)$$

By employing Eqs. (7)-(11), one can easily determine K_p , K_I , K_D , α and β by designing ω_b , $\delta (= 1 + \tan^2 \theta)$ and ε .

Here, the physical image of ε is unclear. Therefore, as well as to [11], using ω_c (crossover frequency of sensitivity function $S(j\omega)$ and complementary sensitivity function $T(j\omega)$), the relationship between $S(j\omega_c)$ and $T(j\omega_c)$ is derived as follows.

$$|S(j\omega_c)| = |T(j\omega_c)|$$

$$\Leftrightarrow \left| \frac{1}{1 + C(j\omega_c)P(j\omega_c)} \right| = \left| \frac{C(j\omega_c)P(j\omega_c)}{1 + C(j\omega_c)P(j\omega_c)} \right|, \quad (12)$$

where $C(j\omega_c) = C_1(j\omega_c) + C_2(j\omega_c)$, the following relationship can be derived.

$$1 = |C(j\omega_c)P(j\omega_c)|$$

$$\approx \frac{K}{\omega_c}. \quad (13)$$

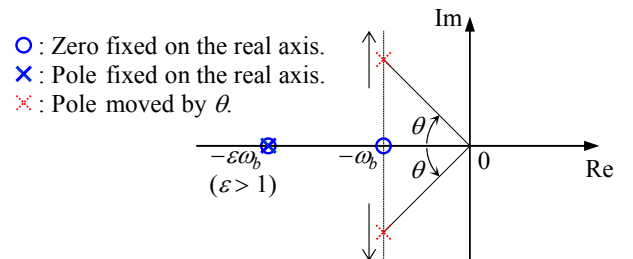


Fig. 3. Pole-zero-assignment in [12].

Therefore, $K \approx \omega_c$. By comparing the coefficient of the term s^2 of the denominator between Eqs. (4) and (6), ε is expressed as follows.

$$\varepsilon \approx \frac{\omega_c}{\omega_b} - 2 \quad (14)$$

From Eq. (14), one can see that ε can be changed by ω_c . Therefore, it is possible to adjust the disturbance suppression characteristic by using ω_c which is easy to have a physical image.

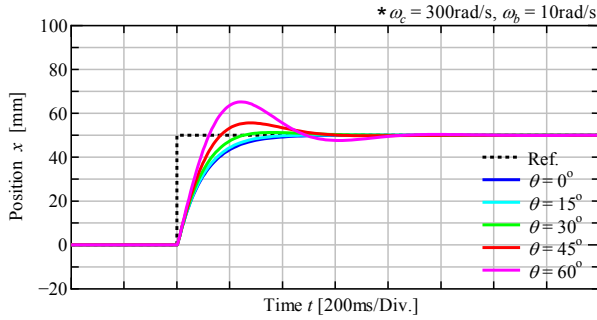
Note that ε is approximately proportional to ω_c when ω_c is sufficiently larger than ω_b .

IV. TRANSIENT RESPONSE CHARACTERISTICS

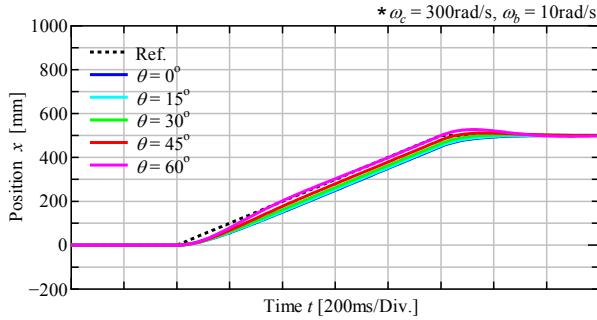
A. Comparison between conventional and proposed methods

Figs. 4 and 5 are simulation results of the characteristics in pole-zero-assignment shown in Fig. 2 and Fig. 3, respectively. Here, $m = 11\text{kg}$, $m_L = 0\text{kg}$, $k = 41.6\text{N/A}$, $\omega_c = 300\text{rad/s}$ and $\omega_b = 10\text{rad/s}$.

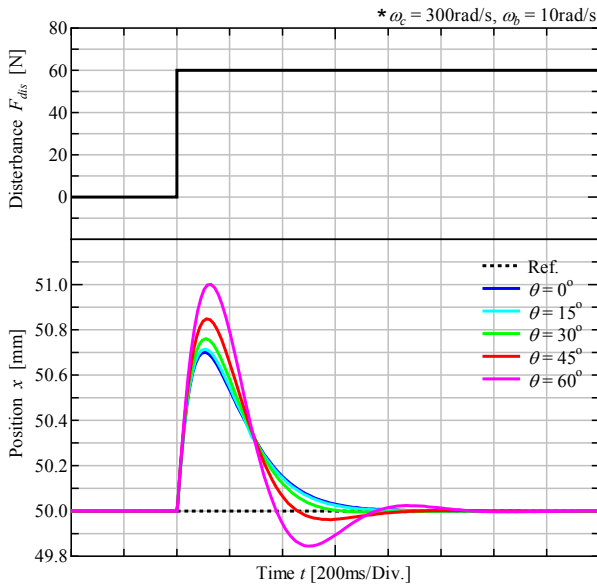
From the results of 50mm step-type position response in Fig. 4 (a) and Fig. 5 (a), it is confirmed that the responses of Fig. 5 (a) are quicker than those of Fig. 4 (a). This is because when the design method based on Fig. 2 is employed, the absolute value of the real part of the



(a) 50mm step-type position responses

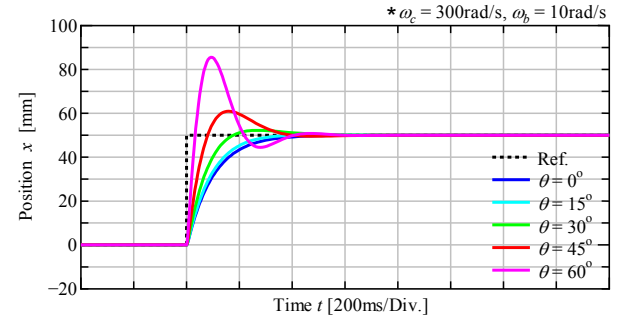


(b) 500mm/s ramp-type position responses

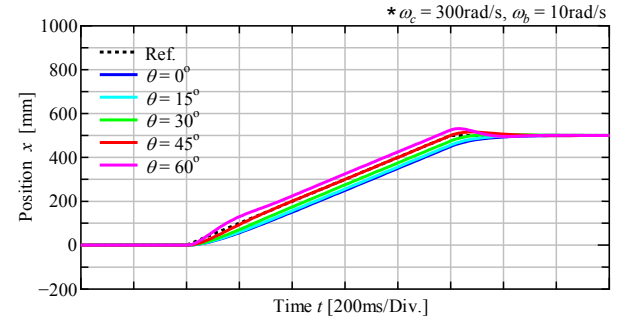


(c) 60N step-type disturbance responses

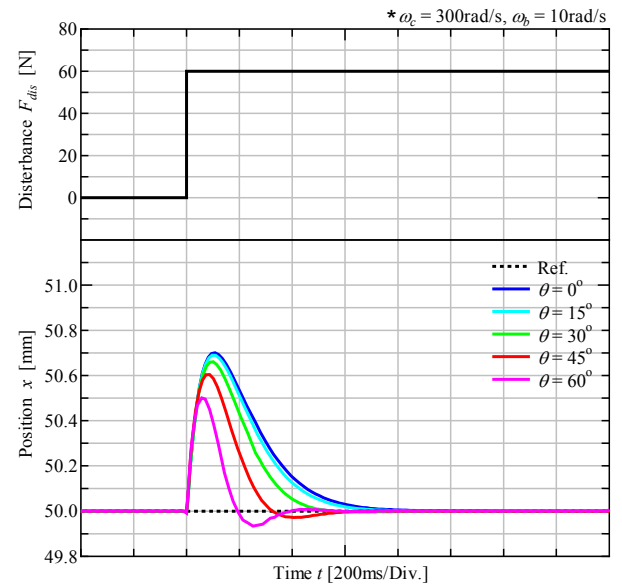
Fig. 4. Simulation results of the pole-zero-assignment in [11].



(a) 50mm step-type position responses



(b) 500mm/s ramp-type position responses



(c) 60N step-type disturbance responses

Fig. 5. Simulation results of the pole-zero-assignment in [12].

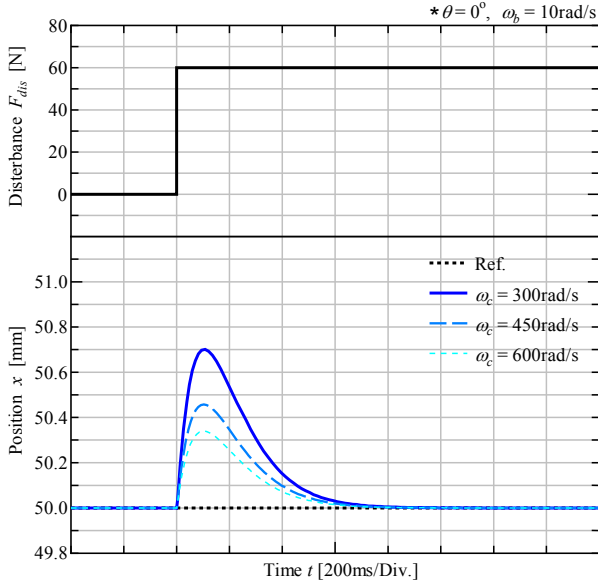
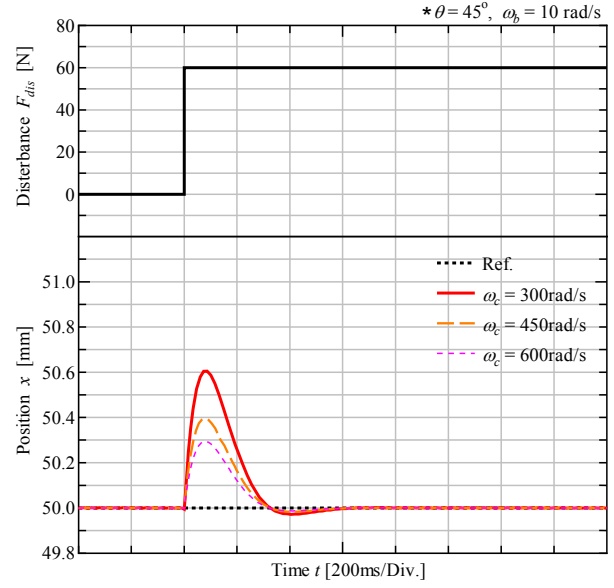

 (a) when positioning drives ($\theta = 0$ degree)

 (b) when tracking drives ($\theta = 45$ degrees)

 Fig. 6. Simulation results of the 60N step-type disturbance response when ω_c varies.

conjugate complex poles is kept in constant. Meanwhile, in case of the design method based on Fig. 3, it decreases as θ increases.

Next, from the results of 500mm/s ramp-type position response in Fig. 4 (b) and Fig. 5 (b), it is observed that in Fig. 4(b) the position error during transient state converges to zero when $\theta = 60$ degrees. Meanwhile in Fig. 5 (b) the error becomes zero when $\theta = 45$ degrees. This difference can be theoretically explained by deriving the condition that the position error for ramp response becomes zero.

Furthermore, from the results of 60N step-type disturbance response in Fig. 4 (c) and Fig. 5 (c), it is found that in the tracking drive ($\theta > 0$), the results of Fig. 5 (c) have higher disturbance suppression characteristics than those of Fig. 4 (c). This is because the absolute value of the real part of the conjugate complex poles of Fig. 2 is larger than that of Fig. 3.

B. Disturbance suppression characteristics

Fig. 6 shows simulation results of 60N step-type disturbance response for several values of ω_c . Fig. 6 (a) shows the characteristics during the positioning drive ($\theta = 0$ degree), and Fig 6 (b) shows the characteristics during the tracking drive ($\theta = 45$ degrees). From the figures, it is confirmed that the larger the ω_c , the stronger the disturbance suppression characteristic. Thus, it can be seen that the disturbance suppression characteristic can be adjusted by changing ω_c .

V. EXPERIMENTAL RESULTS

A. Experimental setup

To verify the validity of the proposed method in [12], the authors carried out actual driving tests of a vector-controlled LSM. Table 1 shows the specifications of the

 TABLE I
SPECIFICATION OF THE TESTED LSM

Ratings	Photo
No. of phases	3
No. of poles	4
Thrust	65.7 N
Current	0.85 A
Speed	2.0 m/s (max)
Frequency	30.3 Hz
Pole pitch	33 mm
Mover mass	11 kg

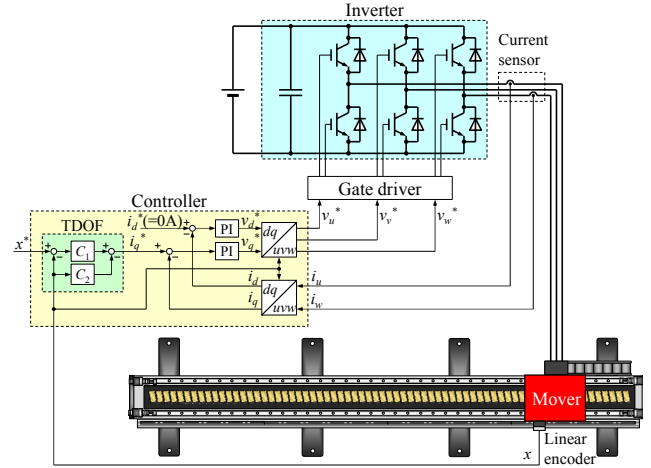
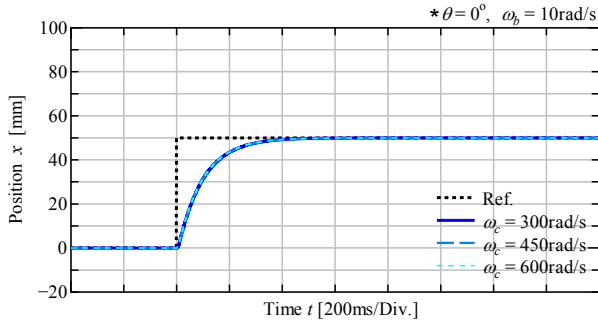


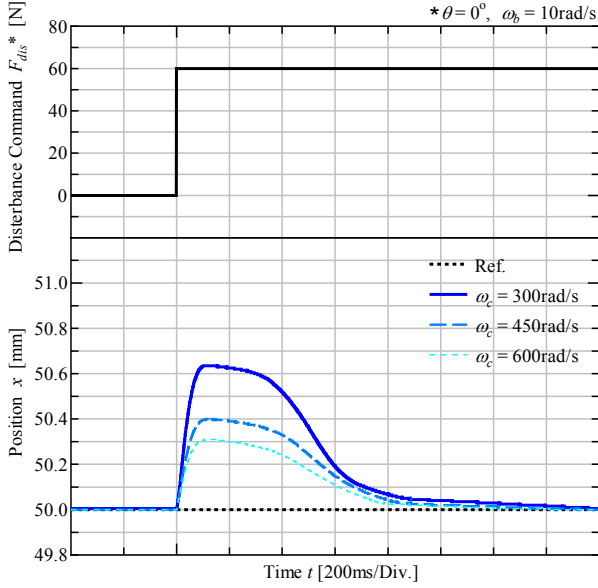
Fig. 7. Experimental setup.

test LSM manufactured by Shinko Electric Co., Ltd..

Fig. 7 shows the experimental setup. The switching frequency of the PWM inverter (Myway Plus Co., Ltd., MWINV-9R122B) is 10kHz. Employing the same condition as the simulation, the authors implement the controller in Fig. 7 on a DSP system (Myway Plus Co., Ltd., Expert III). The actual position data is detected by a linear encoder whose resolution is 5 μ m.



(a) 50mm step-type position responses



(b) 60N step-type disturbance responses

Fig. 8. Experimental results in the positioning drives when ω_c varies.

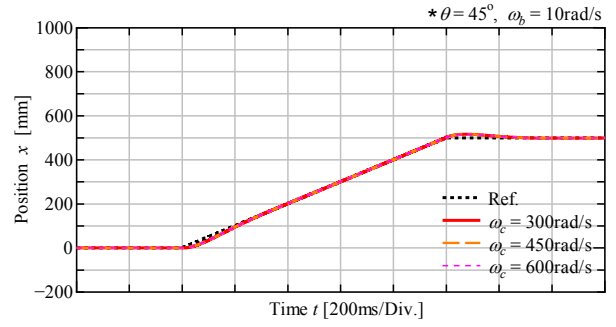
The current and position control periods are $100\mu\text{s}$ and 1ms respectively. The current PI controllers are determined so that the cutoff angular frequency of the current control loop is $2,000\text{rad/s}$. The d-axis current command i_d^* is 0A . Thrust constant is 41.6N/A . The mover mass and loaded-work mass are 11kg and 0kg respectively.

B. Experimental results of positioning drives

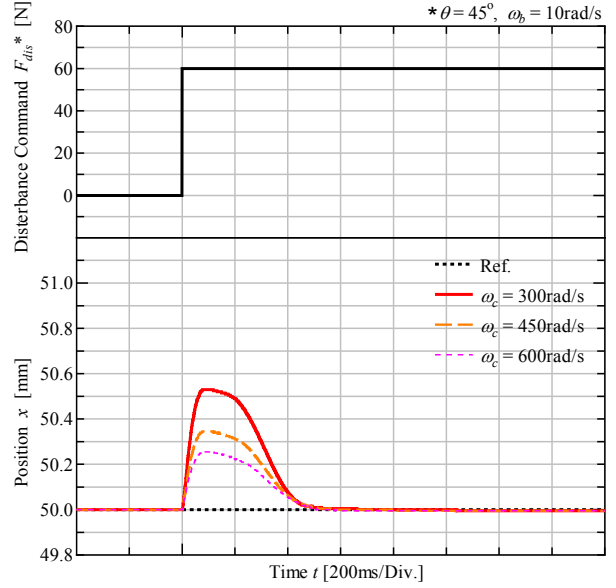
Fig. 8 shows the experimental results of the positioning drives ($\theta = 0$ degree) when ω_c varies. Fig. 8 (a) shows the experimental results of 50mm step-type position response. From Fig. 8 (a), it is confirmed that overshoot-less position responses are obtained regardless of the setting of ω_c . Fig. 8 (b) shows 60N step-type disturbance responses. In Fig. 8 (b), it can be seen that the disturbance suppression characteristic becomes stronger when ω_c increases, as well as in the simulation result of Fig. 6(a).

C. Experimental results of tracking drives

Fig. 9 shows the experimental results of the tracking drives ($\theta = 45$ degrees). Fig. 9 (a) shows the experimental results of 500mm/s ramp-type position response. Fig. 9



(a) 500mm/s ramp-type position responses



(b) 60N step-type disturbance responses

Fig. 9. Experimental results in the tracking drives when ω_c varies.

(a), suggests that position responses without tracking error are realized even when ω_c changes. Fig. 9 (b) shows 60N step-type disturbance responses. In Fig. 9 (b), it can be seen that the disturbance suppression characteristic becomes stronger when ω_c increases, as shown in the simulation result of Fig. 6(b).

VI. CONCLUSIONS AND FUTURE WORK

A simple design method of TDOF PID controller based on pole-zero-assignment for LSMs was presented. The results are summarized as follows.

- (1) In the proposed design method, transient response and disturbance suppression characteristics are adjustable only by the pole angle in the s-plane and crossover frequency of sensitivity and complementary sensitivity function.
- (2) The proposed design method achieves higher response and disturbance suppression characteristics than the conventional design method in the tracking drives.
- (3) Simulation and experimental results clarified the validity of the proposed design method.

Experimental verification under loaded-work mass variation is an important future work.

REFERENCES

- [1] N. Suda, PID control, Asakura Publishing Co., Ltd., 1993 (in Japanese).
- [2] Y. Masuda and Y. Mori, "Design method of 2 degrees of freedom PID control system based on open-loop characteristics," 2006 Annual Meeting Record IEE Japan, Vol. 3, No. 3-051, pp. 65-66, 2006 (in Japanese).
- [3] T. Benjanarasuth, J. Ngamwiwit, N. Komine and Y. Ochiai, "New tuning formulas for two-degree of freedom PID controllers and the application to level control systems," *IEEJ Transaction on Industry Applications*, Vol. 127, No. 5, pp. 485-492, 2007.
- [4] A. Kimura, S. Wakui, M. Hamochi and K. Akatsu, "Application to mechanical system of perfect 2DOF PID and consideration of turning rule," Japan Industry Applications Society Conference 2008, Vol. 2, No. 2-45, pp. II-477-480, 2008 (in Japanese).
- [5] A. Kimura and S. Wakui, "Application of perfect 2 degrees-of-freedom control system to mechanical apparatuses" *Journal of the Japan Society for Precision Engineering*, Vol. 76, No. 1, pp. 116-122, 2010 (in Japanese).
- [6] S. Iang, M. H. Smith and J. Kitchen, "Automatic tuning of two-degree-of-freedom PID control for engine electronic throttle system" SAE Technical Paper, Paper Number: 2010-01-0156, 2010.
- [7] M. Yamamoto, J. Sawaki and K. Matsuse, "Improvement and characteristics of repetitive positioning control of a linear pulse servo motor," *IEEJ Transaction on Industry Applications*, Vol. 119-D, No. 8/9, pp. 1099-1104, 1999 (in Japanese).
- [8] S. Yamamoto, H. Hirahara, A. Tanaka and T. Ara, "Simple Robust Design of Two-Degree-of-Freedom PID Controller for Tracking Drives of Linear Servo Motors," International Conference on Electrical Machines and Systems 2012, LS1B-3, 2012.
- [9] H. Hirahara, B. In, S. Yamamoto, A. Tanaka and T. Ara, "A Design Method of Two-Degree-of-Freedom PID Position Controller Suitable for Tracking Drives of Linear Servo Motors," 2012 IEE-Japan Industry Applications Society Conference, Vol. 3, No. 3-73, pp. III-327-330, 2012 (in Japanese).
- [10] H. Hirahara, A. Tanaka, S. Yamamoto and T. Ara, "A Systematic Design of Two-Degree-of-Freedom PID Controller for both Positioning and Tracking Drives of Linear Servo Motors," International Conference on Electrical Machines and Systems 2016, DS5G-4-6, 2016.
- [11] H. Hirahara, A. Tanaka, S. Yamamoto and T. Ara, "Systematic Design of Two-Degree-of-Freedom PID Controller for Positioning and Tracking Drives of Linear Servo Motors," *IEEJ Journal of Industry Applications*, Vol. 6, No. 6, pp. 443-448, 2017.
- [12] K. Kuroki, H. Hirahara and S. Yamamoto, "Study on Simple Design Method of Two-Degree-of-Freedom PID Position Controller for Linear Servo Motors Based on Pole-Zero Assignment," 2020 Annual Meeting Record IEE Japan, Vol. 5, No. 5-006, pp. 8-9, 2020.